```python
# Step 1: Import libraries and load dataset

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso,
ElasticNet
from sklearn.metrics import mean_squared_error, r2_score

# Load Diabetes dataset
data = load_diabetes()

X = pd.DataFrame(data.data, columns=data.feature_names)
y = data.target

print("Dataset shape:", X.shape)
print(X.head())
```

```
Dataset shape: (442, 10)
        age       sex       bmi        bp        s1        s2
s3  \
0  0.038076  0.050680  0.061696  0.021872 -0.044223 -0.034821 -
0.043401
1 -0.001882 -0.044642 -0.051474 -0.026328 -0.008449 -0.019163
0.074412
2  0.085299  0.050680  0.044451 -0.005670 -0.045599 -0.034194 -
0.032356
3 -0.089063 -0.044642 -0.011595 -0.036656  0.012191  0.024991 -
0.036038
4  0.005383 -0.044642 -0.036385  0.021872  0.003935  0.015596
0.008142

         s4        s5        s6
0 -0.002592  0.019907 -0.017646
1 -0.039493 -0.068332 -0.092204
2 -0.002592  0.002861 -0.025930
3  0.034309  0.022688 -0.009362
4 -0.002592 -0.031988 -0.046641
```

```python
# Step 2: Split the dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

```
print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```

```
Training data: (353, 10)
Testing data: (89, 10)
```

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# Step 3: Linear Regression - Baseline

linear = LinearRegression()

linear.fit(X_train, y_train)

linear_pred = linear.predict(X_test)

# Step 4: Ridge Regression

ridge = Ridge(alpha=1.0)

ridge.fit(X_train, y_train)

ridge_pred = ridge.predict(X_test)

# Step 5: Lasso Regression

lasso = Lasso(alpha=0.1)

lasso.fit(X_train, y_train)

lasso_pred = lasso.predict(X_test)

# Step 6: Elastic Net Regression

elastic = ElasticNet(alpha=0.1, l1_ratio=0.5)

elastic.fit(X_train, y_train)

elastic_pred = elastic.predict(X_test)

# Step 7: Calculate RMSE and R2

models = {
    "Linear Regression": linear_pred,
    "Ridge Regression": ridge_pred,
    "Lasso Regression": lasso_pred,
    "Elastic Net": elastic_pred
}

results = []

for name, prediction in models.items():
    rmse = np.sqrt(mean_squared_error(y_test, prediction))
    r2 = r2_score(y_test, prediction)
```

```
    results.append([name, rmse, r2])

# Display results
results_df = pd.DataFrame(
    results,
    columns=["Model", "RMSE", "R2 Score"]
)

print(results_df)
```

```
                Model       RMSE  R2 Score
0  Linear Regression  53.853446  0.452603
1   Ridge Regression  55.474462  0.419153
2   Lasso Regression  52.897954  0.471855
3        Elastic Net  69.104752  0.098654
```

```
# Step 8: Compare coefficients of all models

coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Linear Regression": linear.coef_,
    "Ridge": ridge.coef_,
    "Lasso": lasso.coef_,
    "Elastic Net": elastic.coef_
})

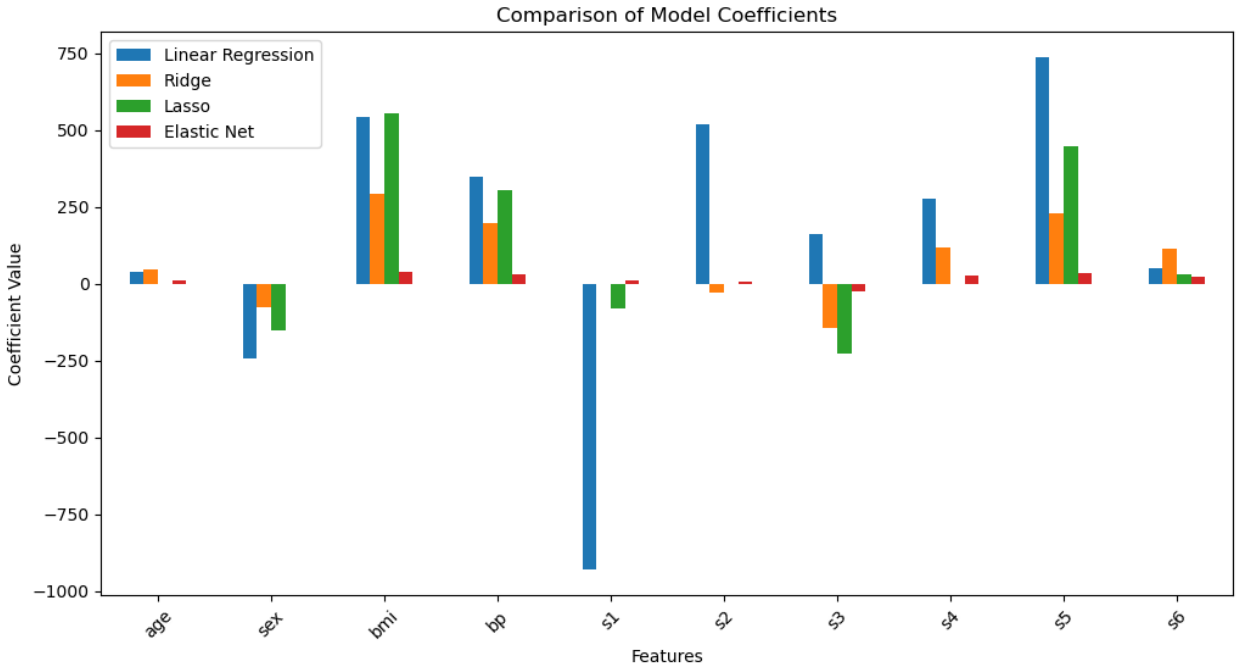
print(coefficients)
```

```
  Feature  Linear Regression       Ridge       Lasso  Elastic Net
0     age          37.904021   45.367377    0.000000    10.830921
1     sex        -241.964362  -76.666086 -152.664779    -0.009514
2     bmi         542.428759  291.338832  552.697775    38.906865
3      bp         347.703844  198.995817  303.365158    28.779233
4      s1        -931.488846   -0.530310  -81.365007    10.372007
5      s2         518.062277  -28.577050   -0.000000     7.191892
6      s3         163.419983 -144.511905 -229.255776   -23.510979
7      s4         275.317902  119.260066    0.000000    25.627160
8      s5         736.198859  230.221608  447.919525    35.073745
9      s6          48.670657  112.149830   29.642617    24.071251
```

```
# Plot coefficients for comparison

coefficients.set_index("Feature").plot(kind="bar", figsize=(12, 6))

plt.title("Comparison of Model Coefficients")
plt.xlabel("Features")
plt.ylabel("Coefficient Value")
plt.xticks(rotation=45)
plt.legend()
plt.show()
```


Comparison of Model Coefficients
Linear Regression
Ridge
Lasso
Elastic Net
Coefficient Value
750
500
250
0
−250
−500
−750
−1000
age
sex
bmi
bp
s1
s2
s3
s4
s5
s6
Features